TRIPATHI STUDY ZONE

JEE MAIN | NEET | AIIMS | KVPY | BITSAT

Mock Test 6

Time : 3 hrs. **Max. Marks : 300**

INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three parts in the question paper consisting of Physics (Q.no.1 to 30), Chemistry (Q.no.31 to 60) and Mathematics (Q. no.61 to 90). Each part is divided into two sections, Section A consists of 20 multiple choice questions & Section B consists of 10 Numerical value answer Questions.
   In Section B, candidates have to attempt **only 5 questions out of 10**.
5. There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for unattempted question. For Section B 4 marks will be awarded for correct answer and zero for marked for each review / unattempted/incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

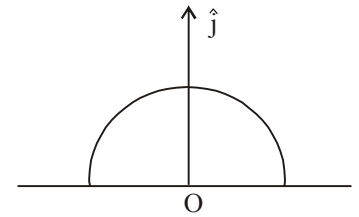
Section - A

1. A 1 kg block attached to a spring vibrates with a frequency of 1 Hz on a frictionless horizontal table. Two springs identical to the original spring are attached in parallel to an 8 kg block placed on the same table. The frequency of vibration of the 8 kg block is :

   (1) $\frac{1}{4}$ Hz (2) $\frac{1}{2\sqrt{2}}$ Hz

   (3) $\frac{1}{2}$ Hz (4) $\sqrt{2}$ Hz

2. A thin semi-circular ring of radius r has a positive charge q distributed uniformly over it. The net field $\vec{E}$ at the centre O is



   (1) $\frac{q}{4\pi^2\varepsilon_0 r^2}\hat{j}$ (2) $-\frac{q}{4\pi^2\varepsilon_0 r^2}\hat{j}$

   (3) $-\frac{q}{2\pi^2\varepsilon_0 r^2}\hat{j}$ (4) $\frac{q}{2\pi^2\varepsilon_0 r^2}\hat{j}$

Space for Rough Work

3. If a bar magnet of magnetic moment 80 units be cut into two halves of equal lengths, the magnetic moment of each half will be

(1) 80 units (2) 40 units

(3) 160 units (4) 20 units

4. Hydrogen $\left({}_1H^1\right)$, Deuterium $\left({}_1H^2\right)$, singly ionised Helium $\left({}_2He^4\right)^+$ and doubly ionised lithium $\left({}_3Li^6\right)^{++}$ all have one electron around the nucleus. Consider an electron transition from n = 2 to n = 1. If the wavelengths of emitted radiation are λ_1, λ_2, λ_3 and λ_4 respectively then approximately which one of the following is correct?

(1) $4\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$

(2) $\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$

(3) $\lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$

(4) $\lambda_1 = 2\lambda_2 = 3\lambda_3 = 4\lambda_4$

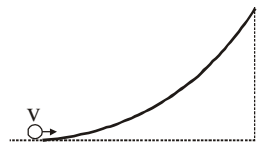
5. A small object of uniform density rolls up a curved surface with an initial velocity v. It reaches up to a maximum height of $\frac{3v^2}{4g}$ with respect to the initial position. The object is

(1) Ring

(2) Solid sphere

(3) Hollow sphere

(4) Disc



6. The amplitude of velocity of a particle acted on by a force F cos ωt along the x direction, is given by $x = \frac{1}{\sqrt{[a\omega^2 - b\omega + c]}}$

where a, b, c, are constants and $b^2 < 4$ ac. For what value of ω does the resonance occur ?

(1) $\omega = \frac{b}{2a}$ (2) $\omega = \frac{b}{a}$

(3) $\omega = c$ (4) $\omega = 0$

7. Three bars each of area of cross - section A and length L are connected in series. The thermal conductivities of their materials are K, 2K, 1.5K. If the temperatures of the external ends of the first and last bar are 200 & 18°C, then the temperatures of both the junctions are

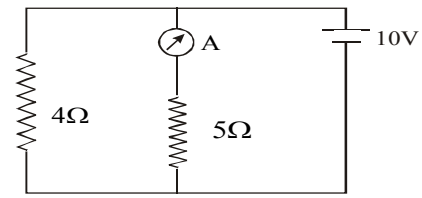
(1) $T_1 = 116°C, T_2 = 74°C$

(2) $T_1 = 120°C, T_2 = 84°C$

(3) $T_1 = 132°C, T_2 = 98°C$

(4) $T_1 = 164°C, T_2 = 62°C$

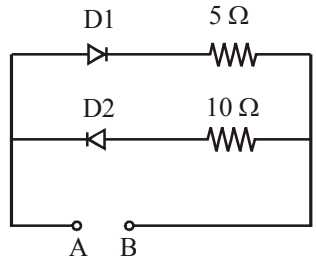
8. In the following diagram the reading of the ammeter is (when the internal resistance of the battery is zero)



(1) $\frac{40}{29}$A (2) $\frac{10}{9}$A

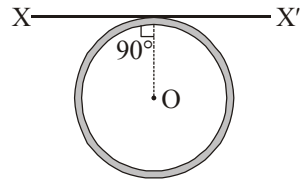
(3) $\frac{5}{3}$A (4) 2 A

9. A 2V battery is connected across AB as shown in the figure. The value of the current supplied by the battery when in one case battery's positive terminal is connected to A and in other case when positive terminal of battery is connected to B will respectively be:



(1) 0.4 A and 0.2 A (2) 0.2 A and 0.4 A
(3) 0.1 A and 0.2 A (4) 0.2 A and 0.1 A

10. A thin wire of length L and uniform linear mass density ρ is bent into a circular loop with centre at O as shown. The moment of inertia of the loop about the axis XX' is



(1) $\frac{\rho L^3}{8\pi^2}$

(2) $\frac{\rho L^3}{16\pi^2}$

(3) $\frac{5\rho L^3}{16\pi^2}$

(4) $\frac{3\rho L^3}{8\pi^2}$

11. In a uniform magnetic field of induction B a wire in the form of a semicircle of radius r rotates about the diameter of the circle with an angular frequency ω. The axis of rotation is perpendicular to the field. If the total resistance of the circuit is R, the mean power generated per period of rotation is

(1) $\frac{(B\pi r\omega)^2}{2R}$ (2) $\frac{(B\pi r^2\omega)^2}{8R}$

(3) $\frac{B\pi r^2\omega}{2R}$ (4) $\frac{(B\pi r\omega^2)^2}{8R}$

12. A steel wire is suspended vertically from a rigid support. When loaded with a weight in air, it extends by l_a and when the weight is immersed completely in water, the extension is reduced to l_w. Then the relatively density of material of the weight is

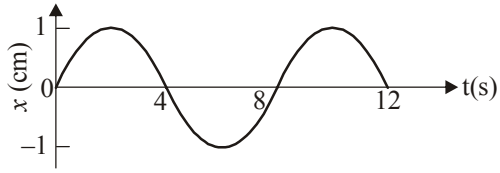
(1) $\frac{l_a}{l_w}$ (2) $\frac{l_a}{l_a - l_w}$

(3) $\frac{l_w}{l_a - l_w}$ (4) $\frac{l_w}{l_a}$

13. An electromagnetic wave of frequency 1×10^{14} hertz is propagating along z-axis. The amplitude of electric field is 4 V/m. If $\varepsilon_0 = 8.8 \times 10^{-12}$ C^2/N-m^2, then average energy density of electric field will be:

(1) 35.2×10^{-10} J/m^3
(2) 35.2×10^{-11} J/m^3
(3) 35.2×10^{-12} J/m^3
(4) 35.2×10^{-13} J/m^3

14. The x-t graph of a particle undergoing simple harmonic motion is shown below. The acceleration of the particle at $t = 4 / 3$ s is



(1) $\frac{\sqrt{3}}{32}\pi^2$ cm/s^2 (2) $\frac{-\pi^2}{32}$ cm/s^2

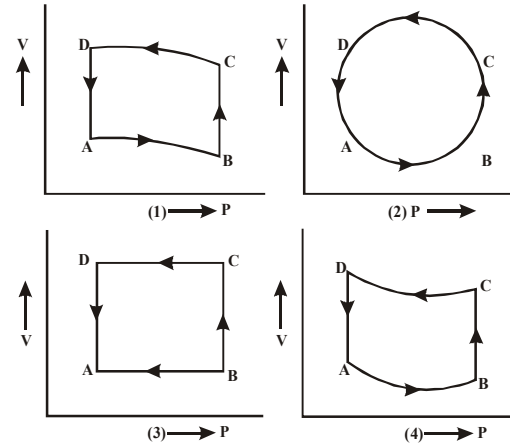
(3) $\frac{\pi^2}{32}$ cm/s^2 (4) $-\frac{\sqrt{3}}{32}\pi^2$ cm/s^2

15. In a Young's double slit experiment with light of wavelength λ the separation of slits is d and distance of screen is D such that D >> d >> λ. If the fringe width is β, the distance from point of maximum intensity to the point where intensity falls to half of maximum intensity on either side is:

(1) $\frac{\beta}{6}$ (2) $\frac{\beta}{3}$

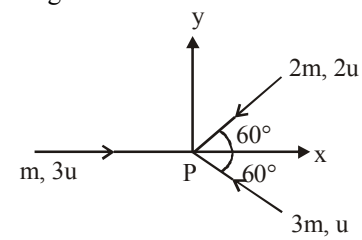
(3) $\frac{\beta}{4}$ (4) $\frac{\beta}{2}$

16. In diagrams (1 to 4), variation of volume with changing pressure is shown. A gas is taken along the path ABCD. The change in internal energy of the gas will be



(1) Positive in all the cases (1) to (4)
(2) Positive in cases (1), (2), (3) but zero in case(4)
(3) Negative in cases (1), (2), (3) but zero in case (4)
(4) Zero in all the cases

17. Three masses m, 2m and 3m are moving in x-y plane with speed 3u, 2u and u respectively as shown in figure. The three masses collide at the same point at P and stick together. The velocity of resulting mass will be:



(1) $\frac{u}{12}\left(\hat{i}+\sqrt{3}\hat{j}\right)$ (2) $\frac{u}{12}\left(\hat{i}-\sqrt{3}\hat{j}\right)$

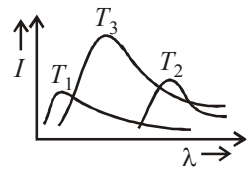
(3) $\frac{u}{12}\left(-\hat{i}+\sqrt{3}\hat{j}\right)$ (4) $\frac{u}{12}\left(-\hat{i}-\sqrt{3}\hat{j}\right)$

18. A satellite of mass m revolves around the earth of radius R at a height x from its surface. If g is the acceleration due to gravity on the surface of the earth, the orbital speed of the satellite is

(1) $\frac{gR^2}{R+x}$ (2) $\frac{gR}{R-x}$

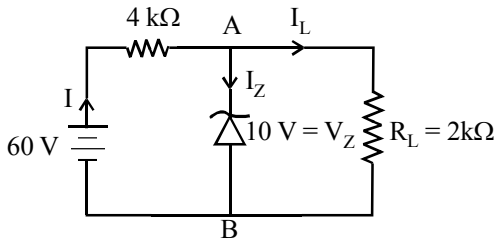
(3) gx (4) $\left(\frac{gR^2}{R+x}\right)^{1/2}$

19. The plots of intensity versus wavelength for three black bodies at temperatures T_1, T_2 and T_3 respectively are as shown. Their temperature are such that



(1) $T_1 > T_2 > T_3$
(2) $T_1 > T_3 > T_2$
(3) $T_2 > T_3 > T_1$
(4) $T_3 > T_2 > T_1$

20. A Zener diode is connected to a battery and a load as shown below:



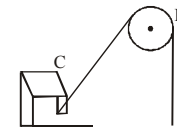
The currents, I, I_Z and I_L are respectively.

(1) 15 mA, 5 mA, 10 mA
(2) 15 mA, 7.5 mA, 7.5 mA
(3) 12.5 mA, 5 mA, 7.5 mA
(4) 12.5 mA, 7.5 mA, 5 mA

Section - B

21. The magnetic field due to a current carrying circular loop of radius 3 cm at a point on the axis at a distance of 4 cm from the centre is 54 μT. What will be its value at the centre of loop (in μT)?

22. One end of a massless rope, which passes over a massless and frictionless pulley P is tied to a hook C while the other end is free. Maximum tension that the rope can bear is 360 N. With what value of maximum safe acceleration (in ms^{-2}) can a man of 60 kg climb on the rope?



23. The velocity of projection of a body is increased by 2%, other factors remaining unchanged, what will be the percentage change in the maximum height attained?

24. The insulation property of air breaks down at $E = 3 \times 10^6$ volts/ metre. The maximum charge that can be given to a sphere of diameter 5 m is approximately _____ coulomb.

25. A compound microscope has an objective and eye-piece as thin lenses of focal lengths 1 cm and 5 cm respectively. The distance between the objective and the eye-piece is 20 cm. The distance at which the object must be placed in front of the objective if the final image is located at 25 cm from the eye-piece, is numerically _____ cm.

Space for Rough Work

26. When light of 3000 Å is incident on sodium chloride, the stopping potential is 1.85 volt and when light of 4000 Å is incident, then the stopping potential becomes 0.82 volt. The threshold wavelength for sodium is _____ Å. ($h = 6.6 \times 10^{-34}$ J-sec.)

27. In four complete revolution of the cap, the distance travelled on the pitch scale is 2 mm. If there are 50 divisions on the circular scale, then the least count of the screw gauge is _____ mm.

28. An electric charge 10^{-3} μ C is placed at the origin (0, 0) of $X - Y$ co-ordinate system. Two points A and B are situated at $(\sqrt{2}, \sqrt{2})$ and (2, 0) respectively. The potential difference between the points A and B will be _____ volts.

29. The Thallium-201 half-life is 74 hours. If the sample has an activity of 80 millicuries initially, what will be the activity (in mci) after 9.25 days ?

30. An engine approaches a hill with a constant speed. When it is at a distance of 0.9 km, it blows a whistle whose echo is heard by the driver after 5 seconds. If the speed of sound in air is 330 m/s, then the speed of the engine (in m/s) is _____ .

CHEMISTRY

Section - A

31. Four species are listed below:

i. HCO_3^- ii. H_3O^+

iii. HSO_4^- iv. HSO_3F

Which one of the following is the correct sequence of their acid strength?

(1) iv < ii < iii < i (2) ii < iii < i < iv

(3) i < iii < ii < iv (4) iii < i < iv < ii

32. Which one of the following pairs is mismatched

(1) Fossil fuel burning - release of CO_2

(2) Nuclear power - radioactive wastes

(3) Solar energy - Greenhouse effect

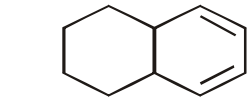
(4) Biomass burning - release of CO_2

33. Which of the following undergoes hydrolysis by S_N1 mechanism :

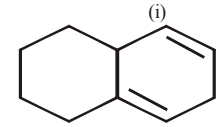
(1) $CH_3CH_2CH_2CH_2Cl$ (2) $CH_3CH_2CH_2Cl$

(3) $CH_3 - O - CH_2Cl$ (4) CH_3Cl

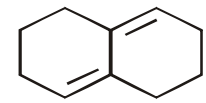
34. Which of the following hydrocarbon can react with maleic anhydride ?



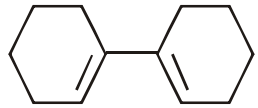
(i)



(ii)



(iii)



(iv)

(1) only (i) (2) (i) and (iv)
(3) (i), (iii) and (iv) (4) all the four

35. The process of the extraction of Au and Ag is based on their solubility in
(1) NH_3 or NH_4OH (2) KCN or NaCN
(3) HCl or HNO_3 (4) H_2SO_4

36. Which of the following compounds having highest and lowest melting point respectively ?
(1) CsF (2) LiF
(3) HCl (4) HF
Correct answer is
(1) 2, 3 (2) 1, 4
(3) 4, 3 (4) 2, 1

37. Silver bromide when dissolve in hypo solution gives complex ..... in which oxidation state of silver is ....
(1) $Na_3[Ag(S_2O_3)_2]$, (I)
(2) $Na_3[Ag(S_2O_3)_3]$, (III)
(3) $Na_3[Ag(S_2O_3)_2]$, (II)
(4) $Na_3[Ag(S_2O_3)_4]$, (I)

38. An organic compound (A) reacts with sodium metal and forms (B). On heating with conc. H_2SO_4, (A) gives diethyl ether, (A) and (B) are
(1) C_2H_5OH and C_2H_5ONa
(2) C_3H_7OH and CH_3ONa
(3) CH_3OH and CH_3ONa
(4) C_4H_9OH and C_4H_9ONa

39. Which one of the following esters cannot undergo Claisen self-condensation?
(1) $CH_3-CH_2-CH_2-CH_2-COOC_2H_5$
(2) $C_6H_5COOC_2H_5$
(3) $C_6H_5CH_2COOC_2H_5$
(4) $C_6H_{11}CH_2COOC_2H_5$

40. In recovery of silver from photographic film, you have decided to dissolve the silver ion with dilute nitric acid. Addition of dilute HCl to precipitate AgCl seems to result in unacceptable losses. You might improve recovery by addition of_____ in the latter step.
(1) $NaNO_3$ (2) NaCl
(3) Ag_2SO_4 (4) sodium acetate

41. One mole of fluorine is reacted with two moles of hot and concentrated KOH. The products formed are KF, H_2O and O_2. The molar ratio of KF, H_2O and O_2 respectively is
(1) 1 : 1 : 2 (2) 2 : 1 : 0.5
(3) 1 : 2 : 1 (4) 2 : 1 : 2

42. In the reaction,

$$CH_2=CH_2 \xrightarrow{HOCl} A \xrightarrow{B} \begin{matrix} CH_2OH \\ | \\ CH_2OH \end{matrix}$$

The molecule 'A' and the reagent 'B' are
(1) H_2C-CH_2 (epoxide, O) and hot water
(2) CH_3CH_2Cl and NaOH
(3) CH_3CH_2OH and H_2SO_4
(4) $\begin{matrix} CH_2-CH_2 \\ | \quad\quad | \\ Cl \quad\quad OH \end{matrix}$ and $NaHCO_3$

Space for Rough Work

43. Alkali metals dissolve in liquid NH_3 then which of the following observations is not true?

(1) H_2 gas is liberated.

(2) Solution turns into blue due to solvated electrons.

(3) It becomes diamagnetic.

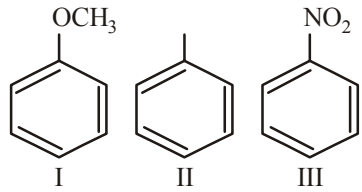
(4) Solution becomes conducting.

44. At very high pressures, the compressibility factor of one mole of a gas is given by :

(1) $1+\frac{Pb}{RT}$ (2) $\frac{Pb}{RT}$

(3) $1-\frac{Pb}{RT}$ (4) $1-\frac{b}{(VRT)}$

45. Among the following compounds (I-III) the correct order of reactivity with electrophile is



(1) II > III > I (2) III < I < II

(3) I > II > III (4) I = II > III

46. One desires to prepare a positively charged sol of silver iodide. This can be achieved by

(1) adding a little $AgNO_3$ solution to KI solution in slight excess.

(2) adding a little KI solution to $AgNO_3$ solution in slight excess.

(3) mixing equal volumes of equimolar solutions of $AgNO_3$ and KI.

(4) none of these.

47. Which of the following is an artificial edible colour?

(1) Melamine (2) Saffron

(3) Carotene (4) Tetrazine

48. Lassaigne's test for nitrogen is positive for which compound?

(1) NH_2OH (2) NH_2NH_2

(3) H_2NCONH_2 (4) All the three

49. It is always advisable not to keep egg yolk or mustard in silver cutlery because

(1) silver reacts with water of egg yolk to form AgOH.

(2) silver reacts with sulphur of egg yolk forming black Ag_2S.

(3) silver reacts with egg yolk forming Ag_2SO_4 which is a poisonous substance.

(4) silver attracts UV light of the atmosphere, thereby spoiling the food.

50. Which of the following complexes will give white precipitate with $BaCl_2(aq)$?

(1) $[Co(NH_3)_4SO_4]NO_2$

(2) $[Cr(NH_3)_4SO_4]Cl$

(3) $[Cr(NH_3)_5Cl]SO_4$

(4) Both (b) & (c)

Section - B

51. Calculate the molality of 1 litre solution of 93% H_2SO_4 (weight/volume). The density of the solution is 1.84 g/mL.
52. Find out the number of stereoisomers obtained by bromination of *trans*-2-butene.
53. The enthalpy of combustion of $H_2(g)$, to give $H_2O(g)$ is –249 kJ mol^{-1} and bond enthalpies of H – H and O = O are 433 kJ mol^{-1} and 492 kJ mol^{-1} respectively. The bond enthalpy of O – H is
54. A certain metal when irradiated to light (v = 3.2× 10^{16} Hz) emits photoelectrons with twice kinetic energy as did photoelectrons when the same metal is irradiated by light (v = 2.0 × 10^{16} Hz) The v_0 (threshold frequency) of metal is $x \times 10^{15}$. Find the value of x.
55. Malachite has the formula $Cu_2CO_3(OH)_2$. What percentage by mass of malachite is copper?
56. The freezing point of a solution, prepared from 1.25 g of a non-electrolyte and 20 g of water, is 271.9 K. If molar depression constant is 1.86 K $mole^{-1}$, find the molar mass of the solute
57. A solution of Ni $(NO_3)_2$ is electrolysed between platinum electrodes using 0.1 Faraday electricity. How many mole of Ni will be deposited at the cathode?
58. In O_2^-, O_2 and O_2^{2-} molecular species. Find the total number of antibonding electrons.
59. x × 10^{21} unit cells are present in a cube-shaped ideal crystal of NaCl of mass 1.00 g. Find the value of x.
[Atomic masses: Na = 23, Cl = 35.5]
60. The electrode potential of Mg^{2+}/Mg electrode, in which conc. of Mg^{2+} is 0.01M, is
($E^o_{Mg^{2+}/Mg} = -2.36V$)

MATHEMATICS

Section - A

61. If $x = \int_0^y \frac{dt}{\sqrt{1+t^2}}$, then $\frac{d^2y}{dx^2}$ is equal to :

(1) y (2) $\sqrt{1+y^2}$

(3) $\frac{x}{\sqrt{1+y^2}}$ (4) y^2

62. The number of real roots of the equation
$1 + a_1 x + a_2 x^2 + ... + a_n x^n = 0$
where $|x| < \frac{1}{3}$ and $|a_n| < 2$, is

(1) n if n is even
(2) 0 for any natural number n
(3) 1 if n is odd
(4) None of these.

63. If $e^{-\frac{\pi}{2}} < \theta < \frac{\pi}{2}$, then

(1) $\cos \log \theta > \log \cos \theta$

(2) $\cos \log \theta < \log \cos \theta$

(3) $\cos \log \theta = \log \cos \theta$

(4) $\cos \log \theta = \frac{2}{3} \log \cos \theta$

64. The value of $\sum_{r=0}^{s} \sum_{\substack{s=1 \\ r \le s}}^{n} {}^{n}C_s \, {}^{s}C_r$ is

(1) $3^n - 1$ (2) $3^n + 1$

(3) 3^n (4) $3(3^n - 1)$

65. Number of ways to distribute 10 distinct balls to 3 persons. One getting 2, 2nd getting 3 and 3rd getting 5 balls is

(1) $\frac{10!}{2!.3!.5!}$ (2) $\frac{10!}{2!.(3!)^2.5!}$

(3) $\frac{10!}{2!.5!}$ (4) 10!

66. The general solution of the differential equation, $\sin 2x \left(\frac{dy}{dx} - \sqrt{\tan x} \right) - y = 0$, is :

(1) $y\sqrt{\tan x} = x + c$

(2) $y\sqrt{\cot x} = \tan x + c$

(3) $y\sqrt{\tan x} = \cot x + c$

(4) $y\sqrt{\cot x} = x + c$

67. In any triangle ABC, if $\cos A = \frac{\sin B}{2 \sin C}$, then

(1) $a = b = c$ (2) $c = a$

(3) $a = b$ (4) $b = c$

68. If ω is a cube root of unity, then a root of the following equation $\begin{vmatrix} x+1 & \omega & \omega^2 \\ \omega & x+\omega^2 & 1 \\ \omega^2 & 1 & x+\omega \end{vmatrix} = 0$ is

(1) $x = 1$ (2) $x = \omega$

(3) $x = \omega^2$ (4) $x = 0$.

69. The lines $\ell x + my + n = 0$, $mx + ny + \ell = 0$ and $nx + \ell y + m = 0$ are concurrent if

(1) $\ell - m - n = 0$

(2) $\ell + m - n = 0$

(3) $\ell - m + n = 0$

(4) $\ell^2 + m^2 + n^2 = \ell m + mn + n\ell$

70. If (x, y) are the co-ordinates of a point in the plane, then $\begin{vmatrix} 3 & 4 & 2 \\ 5 & 8 & 2 \\ x & y & 2 \end{vmatrix} = 0$ represent

(1) a. st. line $\parallel$ to y-axis (2) a st. line $\parallel$ to x-axis

(3) a st. line (4) a circle

71. The equation of the circle whose radius is 5 and which touches the circle $x^2 + y^2 - 2x - 4y - 20 = 0$ at the point (5,5) is

(1) $x^2 + y^2 + 18x + 16y + 120 = 0$

(2) $x^2 + y^2 - 18x - 16y + 120 = 0$

(3) $x^2 + y^2 - 18x + 16y + 120 = 0$

(4) $x^2 + y^2 + 18x - 16y + 120 = 0$

72. The normal at $\left(2,\frac{3}{2}\right)$ to the ellipse, $\frac{x^2}{16}+\frac{y^2}{3}=1$ touches a parabola, whose equation is

(1) $y^2=-104x$ (2) $y^2=14x$

(3) $y^2=26x$ (4) $y^2=-14x$

73. If p : I study and q : I fail

Then negation of 'I study or I fail' is

(1) I do not study and I do not fail

(2) I do not study or I do not fail

(3) Either I study and I do not rail or I study and I do not fail

(4) I study and I do not fail

74. The general solution of the trigonometric equation $\sin x - \cos x = 1$ is given by

(1) $x = 2n\pi, n \in I$

(2) $x = n\pi + (-1)^n\frac{\pi}{4}+\frac{\pi}{4}, n \in I$

(3) $x = 2n\pi + \frac{\pi}{2}, n \in I$

(4) $x = n\frac{\pi}{2}, n \in I$

75. If $\vec{A}=\hat{i}+\hat{j}+\hat{k}$, $\vec{B}=4\hat{i}+3\hat{j}+4\hat{k}$ and $\vec{C}=\hat{i}+\alpha\hat{j}+\beta\hat{k}$ are linearly independent vectors and $|\vec{C}|=\sqrt{3}$, then

(1) $\alpha=1,\beta=-1$ (2) $\alpha=1,\beta=\pm1$

(3) $\alpha=-1,\beta=\pm1$ (4) $\alpha=\pm1,\beta=1$

76. p : Every quadratic equation has one real root and q : Every quadratic equation has two real roots, then truth value of p and q are

(1) p is true and q is false

(2) p is false and q is true

(3) p and q both true

(4) p and q both false

77. Let A, B and C be $n \times n$ matrices. Which one of the following is a correct statement?

(1) If $AB = AC$, then $B = C$

(2) If $A^3+2A^2+3A+5I=0$; then A is invertible.

(3) If $A^2=0$, then $A=0$

(4) None of these

78. If for a real number y, [y] is the greatest integer less than or equal to y, then the value of the integral $\int_{\pi/2}^{3\pi/2}[2\sin x]dx$ is

(1) $-\pi$ (2) π

(3) $\pi/2$ (4) $-\pi/2$

79. If $f(x)=ax^2+b, b\neq0, x\leq1$;

$=bx^2+ax+c, x>1,$

then f(x) is continuous and differentiable at x = 1 if

(1) $c=0, a=2b$ (2) a = b, c arbitrary

(3) $a=b, c=0$ (4) $a=b, c\neq0$

80. The values of the parameter a such that the roots α, β of the equation $2x^2 + 6x + a = 0$ satisfy the inequality $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} < 2$ are

(1) $a > 0$ (2) $a < 9/2$

(3) $a < 0$ or $a > 9/2$ (4) None of these

Section - B

81. It has been found that A and B play a game 12 times, A wins 6 times, B wins 4 times and they draw twice. A and B take part in a series of 3 games. The probability that they will win alternately is _____ .

82. If period of the function $\left|\sin^3\frac{x}{2}\right| + \left|\cos^5\frac{x}{5}\right| = k\pi$, then value of x is _____ .

83. If the least difference between the roots, in the first quadrant $\left(0 \le x \le \frac{\pi}{2}\right)$, of the equation $4\cos x(2 - 3\sin^2 x) + (\cos 2x + 1) = 0$ is $\frac{\pi}{n}$, then value of n is _____ .

84. If $[x]$ denotes the greatest integer $\le x$ and $\lim\limits_{x\to 3} \frac{1}{n^3}([1^2 x] + [2^2 x] + [3^2 x] + \ldots\ldots + [n^2 x]) = \frac{\pi}{n}$, then value of n is _____ .

85. For natural numbers m, n if $(1-y)^m(1+y)^n = 1 + a_1 y + a_2 y^2 + \ldots$ and $a_1 = a_2 = 10$, then value of $m + n$ is _____ .

86. The value of expression

$$3\left[\sin^4\left(\frac{3\pi}{2} - \alpha\right) + \sin^4(3\pi + \alpha)\right]$$

$$-2\left[\sin^6\left(\frac{\pi}{2} + \alpha\right) + \sin^6(5\pi - \alpha)\right] \text{ is } _____ .$$

87. The value of $[\vec{A} - \vec{B}, \vec{B} - \vec{C}, \vec{C} - \vec{A}]$ where $|\vec{A}| = 1$, $|\vec{B}| = 2$ and $|\vec{C}| = 3$ is _____ .

88. If $f(x) = \frac{2}{x-3}$, $g(x) = \frac{x-3}{x+4}$

$h(x) = -\frac{2(2x+1)}{x^2 + x - 12}$ and

$\lim\limits_{x \to 3} [f(x) + g(x) + h(x)] = k$ then value of $|k|$ is _____ .

89. If P and Q be two given points on the curve $y = x + \frac{1}{x}$ such that $\overrightarrow{OP}.\hat{i} = 1$ and $\overrightarrow{OQ}.\hat{i} = -1$ where $\hat{i}$ is a unit vector along the x–axis and the length of vector $2\overrightarrow{OP} + 3\overrightarrow{OQ} = n\sqrt{5}$, then value of n is _____ .

90. The area bounded by parabola $y^2 = x$, st. line $y = 4$ and y-axis is _____ .

Space for Rough Work